Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/12/2008
PI Name:	Farouk, Bakhtier Ph.D.		
Project Title:	Thermoacoustic Convection and Transport in Gases and Near-Critical Fluids under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-2816	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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COI Name (Institution):	Oran, Elaine (Naval Research Laboratory)		
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Task Description:	The research project is aimed at studying thermoacoustic and buoyancy-driven transport in gases and in fluids near their critical points at normal and reduced gravity environments. It is envisioned that supercritical fluid reactors will play a vital role in the development of 'water recovery' and waste treatment systems in long duration manned space missions. The present four-year project (start date: December 8, 2003) combines computational and experimental studies of thermoacoustic waves and thermoacoustic convection in gases and in near-critical fluids. In the first two years of the project, significant advances have been made in both the model development and the experimental design and facility development. Large-scale detailed computational models were developed and tested to simulate, and understand the thermoacoustically driven flows and transport in gases and the generation and propagation of thermoacoustic waves in supercritical carbon dioxide. Both normal-gravity and reduced-gravity experiments are planned to validate the key features of the simulations. For normal-gravity experiments in gases and in supercritical fluids, two experimental facilities have been designed and built at Drexel University. We extended our computational fluid dynamic models (for predicting thermoacoustic waves in supercritical fluids) by incorporating the NIST Reference Database 12 for supercritical carbon dioxide properties. In the case of low-heat-diffusing, near-critical fluids, it has been recently shown that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. The piston-effect has been successfully captured in our simulations using the models developed. Normal gravity experiments were conducted for the generation and characterization of thermoacoustic waves in gases. Preliminary data agree well with model predictions. An experimental setup to generate and characterize the thermoacoustic waves in supercritical carbon dioxide has been designed and fabricated. Studies will be carried out under both normal and reduced gravity environments. Regarding reduced-gravity experiments at the NASA Glenn 2.2 s drop tower, we visited the facilities in August 2005. Modified experimental setups are now being designed that will satisfy the drop tower capabilities and guidelines. The present research is aimed at developing advanced computational tools to analyze interactions between thermoacoustic waves and viscous fluids in several problems of significant relevance to microgravity conditions. The specific objectives of the proposed research are the following: • Improve and refine a high-order computational model for the simulation of flow and heat transfer in gases and near-critical fluids induced by thermoacoustic waves • Perform direct numerical simulation of thermoacoustically driven gas flows in 2-D and 3-D enclosures • Completely characterize the behavior of the thermoacoustic waves (generation, propagation, interaction with solid walls and eventual dissipation) in gases and nar-critical fluids • Perform normal-gravity experiments for thermoacoustic waves in gases for model validation • Perform reduced-gravity experiments (short-time) in drop-towers or KC 135 aircraft for isolating buoyancy effects from thermoacoustic wave generated flows in gases • Select appropriate equation of state for near-critical fluids and incorporate the equation in the numerical model for flow simulation in such fluids • Perform direct numerical simulation of the piston effect in near-critical fluids in enclosures • Execute model for predicting the 'thermoacoustic' effects in near-critical fluids with and without gravitational effects • Perform normal-gravity experiments with near-critical CO2 in carefully designed experiments (which reduce gravitational effects) • Develop guidelines for the use of the mathematical models as a design and optimization tool for property and transport characterization in supercritical fluids and supercritical reactors. Successful completion of the research tasks will provide us important information regarding the design and operation of supercritical reactors. Such reactors are envisioned for waste management and water recovery in long duration manned space missions. The effects of thermoacoustic waves on transport and mixing in microgravity environments in such reactors willbe significant. The present research will be useful for the design of supercritical carbon dioxide reactors used in the food and pharmaceutical industries and for supercritical water reactors for waste management. The presence of gravitational field leads to both practical and fundamental limits of the resolution in critical phenomena experiments in fluids near the gas-liquid critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. Near the critical point, fluids exhibit anomalies in the transport of heat such that convection and buoyancy phenomena appear for even minute temperature gradients. In the case of low-heat-diffusing, near-critical fluids, it has been shown recently that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. A significant question regarding the 'piston effect' is whether the fast heat transport is due to the strong buoyant convection induced by normal gravity in near-critical fluids or if it is due to the high compressibility of such fluids. Microgravity environment is thus ideal for studying such processes for fluids near the critical point where new fluid physics can be discovered.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Supercritical fluids exhibit a number of specific properties like large density, low viscosity, low thermal diffusivity. In addition, their isothermal compressibility and thermal expansion can become very large, especially when they approach the critical point. The highly variable properties of near-critical fluids make them very attractive for studying many interesting phenomena that hold for all fluids because of the critical universality. Fluids in their supercritical state are increasingly used by the food and waste management industry for their solubilization properties (e.g. supercritical carbon dioxide), as host of "cold" combustion (e.g. supercritical water), in energetics (supercritical thermal or nuclear power plants) and in astronautics (e.g. storage of cryogenic fluids). It is particularly challenging to understand the behavior, flow and transport in a fluid that is close to its critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous

	distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. The research results will provide valuable insights to the transport processes in supercritical reactors - whether they are used on earth or long duration manned space missions.
Task Progress:	During the period December 2006 - November 2007 , we accomplished the following: 1. Developed a test facility for producing and measuring thermoacoustic waves in a gas-filled circular tube. At one end of the tube, a thin metallic foil is positioned. The foil is heated rapidly via an electrical circuit with capacitor banks. 2. Evaluated and tested two pressure transducers (Brüel & Kjær model 4193 condenser microphone and an Endevco model 8507C-1 piezoresistive pressure transducer). 3. Constructed and tested a high pressure test facility for characterizing thermoacoustic waves in supercritical carbon dioxide. 4. Developed and tested a numerical model for studying thermoacoustic waves in gases in enclosures
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Lin Y, Farouk B. "Experimental and Numerical Studies of Thermally Induced Acoustic Waves in a Closed Enclosure." Journal of Thermophysics and Heat Transfer. in press, November 2007. , Nov-2007
Articles in Peer-reviewed Journals	Lin Y, Farouk B. "Heat transfer in a rectangular chamber with differentially heated horizontal walls: Effects of a vibrating sidewall." Int. J. Heat Mass Transfer (in press), November 2007. , Nov-2007
Articles in Peer-reviewed Journals	Lin Y, Farouk B, Oran ES. "Interactions of thermally induced acoustic waves with buoyancy induced flows in rectangular enclosures." Int. J. Heat Mass Transfer (in press), November 2007. , Nov-2007
Articles in Peer-reviewed Journals	Lin Y, Farouk B, Oran ES. "Flows induced by thermoacoustic waves: Effects of gravity." Journal of Thermophysics and Heat Transfer, 2006 Jul-Sep;20(3):376-83. , Jul-2006
Dissertations and Theses	Lin Y. "Acoustic Wave Induced Convection and Transport in Gases under Normal and Microgravity Conditions." Ph. D. Dissertation, Drexel University, June 2007. , Jun-2007
Papers from Meeting Proceedings	Lin Y, Farouk B. "Experimental and Numerical Studies of Thermally Induced Acoustic Waves in a Closed Enclosure." ASME-JSME Thermal Engineering Summer Heat Transfer Conference, July 2007. ASME-JSME Thermal Engineering Summer Heat Transfer Conference, July 2007. Paper HT2007-32011. , Jul-2007